Summarise Traits Within a Taxonomic Group

Description

Resolves the taxonomic group containing 'focal_taxon', collects records for taxa in that group and 'trait_domain', and returns per-taxon summary statistics sorted by 'median'.

Usage

```
summarise_taxonomic_group_traits(
  data_trait_records,
  data_taxon_classification,
  focal_taxon,
  trait_domain,
  taxonomic_rank = "family",
  verbose = TRUE
)
```

Arguments

`data_trait_records`	A data frame of raw trait observations. Must contain columns 'taxon_name' (character), 'trait_domain_name' (character), and 'trait_value' (numeric).
`data_taxon_classification`	A data frame mapping taxon names to taxonomic ranks. Must contain 'sel_name' and the column named by 'taxonomic_rank'.
`focal_taxon`	Character scalar. Name of the focal taxon to look up.
`trait_domain`	Character scalar. Trait domain to summarise (matched against 'trait_domain_name' in 'data_trait_records').
`taxonomic_rank`	Character scalar. Name of the taxonomic rank column in 'data_taxon_classification' to use for grouping (e.g. '"family"', '"genus"', '"order"'). Defaults to '"family"'.

verbose	Logical. If 'TRUE' (default), progress messages are printed to the console via 'cli'.

Details

1. The function looks up 'focal_taxon' in 'data_taxon_classification' to resolve 'taxonomic_rank'. 2. All taxa sharing that rank value are identified. 3. 'data_trait_records' is filtered to those taxa and 'trait_domain', then grouped by 'taxon_name'. 4. If the rank value cannot be resolved (taxon absent from classification or the rank column is 'NA'), only the focal taxon's own records are summarised.

Value

A tibble with one row per taxon that has data for 'trait_domain' within the same 'taxonomic_rank' group as 'focal_taxon'. Columns: 'taxon_name', 'n_records', 'minimum', 'lower_quartile', 'median', 'mean', 'upper_quartile', and 'maximum'. Rows are sorted ascending by 'median'. If the focal taxon or rank value cannot be resolved, only records for 'focal_taxon' are summarised.

See Also

[write_trait_quality_control_report()], [plot_focal_trait_distribution()]

Examples

```
## Not run:

data_aux_classification <-

  readr::read_csv(

    here::here("Data/Input/aux_classification_table.csv"),

    show_col_types = FALSE

  )


data_taxonomic_trait_summary <-

  summarise_taxonomic_group_traits(

    data_trait_records = data_traits_raw,

    data_taxon_classification = data_aux_classification,

    focal_taxon = "Anacyclus clavatus",

    trait_domain = "Leaf Area"

  )


## End(Not run)
```